PAPER

Boost-range scaling of the fluctuations of the Benincasa–Dowker action and its consequences for Everpresent Λ

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Abstract

Everpresent Λ is a proposal inspired by causal sets in which the cosmological constant fluctuates as $\delta\Lambda \propto V^{-1/2}$, with V the spacetime volume of the observer's past, an exponent inherited from the Poisson fluctuations of the number of elements. Moradi, Yazdi and Zilhão asked whether these fluctuations could instead be sourced by the Benincasa–Dowker (BD) action, and gave a formalism for its variance without evaluating it. We evaluate it analytically and with Poisson sprinklings in four dimensions. The leading term, the Poisson noise in the number of nearly null intervals, grows with the range of Lorentz boosts that fits inside the region and scales exactly as $(\rho V)^{2-2/d}$. In causal diamonds, past light cones and past cones of flat FLRW spacetimes reaching the big bang, the remaining terms of the local action are a constant fraction of it over the simulated range ($\rho V \leq 25\,600$), so that $\Delta S \propto (\rho V)^{3/4}$ and $\delta\Lambda \propto V^{-1/4}$ there, rather than the Poisson exponents $1/2$ and $-1/2$; the geometry only fixes the amplitude. For the smeared action this holds only within a smaller range, and the physically relevant limit of strong smearing remains open. Reading the action fluctuation as a fluctuation of the cosmological term, and assuming that the exponent and the temporal correlation measured for the local action carry over to the strongly smeared one, we confront stochastic $\Lambda(t)$ histories with Pantheon+SH0ES supernovae, DESI DR2 baryon acoustic oscillations and Planck distance priors. With the measured correlation, none of 10^4 histories fits as well as ΛCDM ($\Delta\chi^2_{\text{min}} = +74$ with all three probes, $+23$ without the CMB). Within a wider family of dynamics, rare good histories appear only when Λ stays close to a slowly wandering constant.

1 Introduction

If spacetime is a causal set, the cosmological constant may fluctuate. Causal set theory replaces the spacetime continuum by a locally finite partially ordered set, whose elements are related by the causal order and whose number counts spacetime volume [1, 2]; a heuristic argument by Sorkin [3, 4], made precise in the Everpresent Λ model [5, 6, 7, 8], connects this discreteness with Λ. In unimodular gravity Λ is conjugate to the spacetime volume V; if V is only defined up to the Poisson fluctuation $\delta N = \sqrt{N}$ of the number of elements, then Λ fluctuates by $\delta\Lambda \sim V^{-1/2} \sim H^2$. The resulting dark energy density tracks the critical density at all epochs, which would explain why Ω_Λ is of order one today without fine tuning.

The exponent $-1/2$ is essential to the model. Write the fluctuation of the action as $\Delta S \propto V^p$ (this exponent p is used throughout). Then $\delta\Lambda \propto V^{p-1}$ and, since $V \sim H^{-4}$, $\delta\Lambda/H^2 \propto H^{2-4p}$: only $p = 1/2$ gives an *everpresent* Ω_Λ. In the phenomenological realisation of Das, Nasiri and Yazdi (Model 1 of [7]), this exponent is imposed by construction: each causal-set element contributes an independent random variable to the action term ΛV. The authors note that "the true causal set mechanism that gives rise to Everpresent Λ is not fully understood" [7].

A natural candidate for a mechanism is the causal-set action itself. The Benincasa–Dowker (BD) action [9, 10] is a discretisation of the Einstein–Hilbert action, built from counts of intervals, whose mean reproduces the Einstein–Hilbert action in the continuum limit, up to boundary terms [11, 12]. Moradi, Yazdi and Zilhão (MYZ) [13] developed a formalism for the variance of this action under Poisson sprinkling and raised the question of whether its fluctuations could be the origin of Everpresent Λ. They expressed the variance as a set of integrals over pairs of intervals but did not evaluate them.

Fluctuations are a known issue for causal-set operators. The fluctuations of the local causal-set d'Alembertian at a point grow with the density, which motivated the smeared, non-local operators [14, 9]; simulations indicate that the fluctuations of the smeared operator die away at fixed non-locality scale [9, 15], and the stability of the continuum operators has been studied in [16]. For the action of a region, as opposed to the operator at a point, Benincasa, Dowker and Schmitzer called the behaviour of the fluctuations at large density "a major open question" [17]: preliminary two-dimensional simulations showed them growing with the density [18], and the authors suggested that a mesoscale might be needed to tame them. To our knowledge, the variance and its scaling with the number of elements have not been computed beyond these preliminary numerics.

In this paper we compute them and follow the consequences through to cosmological data. Our main results are:

1. The *same-pair* term of the variance, the Poisson noise in the number of nearly null intervals, grows asymptotically as $(\rho V)^{2-2/d}$, with a prefactor set by a boost-range integral G of the region that we compute in closed form or by quadrature. We derive the leading correction, of relative order $(\rho V)^{-1/2}\ln \rho V$ in four dimensions, and verify both against exact means up to $\rho V = 10^8$ (section 3 and appendix C).

2. In all the regions we simulate (causal diamonds, truncated past light cones and past cones of flat FLRW spacetimes reaching the big bang), the remaining correlation terms are a constant fraction of the same-pair term within the measured range ($\rho V \leq 25\,600$) for the local action (section 4). The exponent of ΔS is compatible with 3/4 in four dimensions, not with the Poisson value 1/2. For the smeared action this holds within a smaller range of $\epsilon\rho V$, with a weak residual trend for the strongest smearing.

3. If the action fluctuation is read as a fluctuation of the cosmological term, $\delta\Lambda \propto V^{-1/4}$ (section 5). Only the strongly smeared action has an acceptable amplitude. We therefore assume that two quantities measured for the local action, the exponent and the temporal correlation along nested past cones, carry over to the smeared one, and on that assumption we confront stochastic $\Lambda(t)$ histories with supernova, BAO and CMB data. Over the measured range of the memory these histories are excluded, as is Model 1 in our implementation; within a wider family of dynamics, good histories are rare and occur only for a Λ that barely departs from a slowly wandering constant (section 6).

The mechanism behind the first result is well known in another guise: the non-compactness of the Lorentz group makes the number of links, or of intervals of any fixed small cardinality, grow without bound in a sprinkled causal set [19], and the mean abundances of small intervals in flat regions are known exactly [20]. The same non-compact rapidity range biases the statistics of links [21]. What is new here is its role in the *fluctuations* of the action: we show that it controls their scaling, give the prefactor in closed form as a boost-range integral G valid for any region, including FLRW past cones, quantify the amplitude in the regions relevant for cosmology, and follow the consequences for a fluctuating cosmological constant. In two dimensions this gives the quantitative answer to the question of [17]: the fluctuations grow as $\sqrt{\rho V \ln \rho V}$, and smearing reduces their amplitude but not their growth.

2 The BD action and its variance

This section fixes notation, recalls the MYZ decomposition of the variance and explains why we do not evaluate it in full.

2.1 Sprinklings and the action

A Poisson sprinkling of density ρ into a region M of a d-dimensional Lorentzian manifold produces a causal set whose number of elements N is a Poisson variable with mean ρV, $V = \mathrm{vol}(M)$. For a related pair $y \prec x$ let $n(y,x)$ be the number of elements strictly between them, and let N_k be the number of pairs with $n = k$. We use the BD action in the form of a sum over related pairs,

$$S = a_N N + \sum_{y\prec x} f\big(n(y,x)\big)\,, \tag{1}$$

which is the form analysed by MYZ [13]. For the *local* action, $f(n) = a_n$ for $n < n_d$ and $f = 0$ otherwise, with

$$d=2: \quad (a_N; a_0, a_1, a_2) = (1; -2, 4, -2), \qquad d=4: \quad (a_N; a_0, \ldots, a_3) = \frac{4}{\sqrt{6}}(1; -1, 9, -16, 8). \tag{2}$$

In four dimensions these are the coefficients of [9], normalised so that the mean action tends to $\frac{1}{2\ell^2}\int R\,\mathrm{d}V$ (plus boundary terms [11, 12, 22]), with $\ell = \rho^{-1/4}$. In two dimensions the coefficients are those of [17], for which the mean action of a flat causal diamond tends to 1, its Euler characteristic, up to corrections that vanish as $\rho \to \infty$ [17]. (The action of MYZ, their equation (4.1) evaluated at $\phi = -2\ell^2/\alpha_d$, is $-\sqrt{6}/2$ times ours in four dimensions; this multiplies the variance by 3/2 and leaves every ratio and exponent of this paper unchanged.) The *smeared* action of Sorkin [14] and Dowker and Glaser [10, 23] introduces a non-locality parameter $\epsilon \in (0, 1]$, with $\epsilon = (\ell/\ell_k)^d$ for a non-locality length ℓ_k. It has the same form (1) with

$$a_N = \epsilon^{2/d} a_N^{(1)}, \qquad f_\epsilon(n) = \epsilon^{1+2/d} \sum_{k<n_d} a_k^{(1)} \binom{n}{k} \epsilon^k (1-\epsilon)^{n-k}, \tag{3}$$

where $a^{(1)}$ are the local coefficients (2); the overall factor $\epsilon^{2/d}$ comes from $\ell_\epsilon^2 = \ell^2/\epsilon^{2/d}$, and the additional ϵ in f_ϵ from the normalisation of the smeared operator [10]. In four dimensions, expanding the binomials gives $f_\epsilon(n) = -\frac{4}{\sqrt{6}}\epsilon^{3/2}(1-\epsilon)^n\big[1 - \frac{9\epsilon n}{1-\epsilon} + \frac{8\epsilon^2 n(n-1)}{(1-\epsilon)^2} - \frac{4\epsilon^3 n(n-1)(n-2)}{3(1-\epsilon)^3}\big]$, the weight of the smeared d'Alembertian of [9]. For $\epsilon = 1$ the local action is recovered. The binomial weight is the probability that a pair with n elements in between keeps exactly k of them after thinning each with probability ϵ, so the smeared action can be written with "thinned" abundances $\tilde{N}_k = \sum_{y\prec x}\binom{n}{k}\epsilon^k(1-\epsilon)^{n-k}$, and the MYZ formalism applies to them with minor modifications (appendix B).

2.2 Means and the Mecke formula

All the mean values we need follow from the Mecke formula for Poisson processes [24]. For any function h,

$$\Big\langle \sum_{y\prec x} h\big(n(y,x)\big)\Big\rangle = \rho^2 \int_M \mathrm{d}V_x \int_{M\cap J^-(x)} \mathrm{d}V_y\; \mathbb{E}\,h\big(\mathrm{Poisson}(\rho\,\mathrm{vol}\,I(y,x))\big), \tag{4}$$

where $I(y,x)$ is the causal interval. In a flat causal diamond the inner integral depends on x only through the volume of its past within the diamond, and (4) reduces to two nested one-dimensional quadratures. We also need the covariance of N with the pair sum $F = \sum_{y\prec x} f(n)$. Differentiating (4) with respect to the intensity gives (appendix A)

$$\mathrm{Cov}(N, F) = \rho\,\frac{\mathrm{d}\langle F\rangle}{\mathrm{d}\rho}\,, \tag{5}$$

which for the local action reduces to the identity $\mathrm{Cov}(N, N_k) = (k+2)L_k - (k+1)L_{k+1}$ of MYZ, with $L_k = \langle N_k\rangle$.

2.3 The MYZ decomposition

The variance of (1) is

$$(\Delta S)^2 = a_N^2\,\rho V + 2a_N\,\mathrm{Cov}(N, F) + \mathrm{Var}(F)\,, \tag{6}$$

and MYZ decompose $\mathrm{Var}(F)$ according to how the two pairs (y_1, x_1), (y_2, x_2) in F^2 share their endpoints:

$$\mathrm{Var}(F) = J^1 + J^2 + J^3 + 2J^4 + J_{\mathrm{c}}^6\,. \tag{7}$$

Here $J^1 = \langle\sum_{y\prec x} f(n)^2\rangle$ is the *same-pair* term; J^2 and J^3 collect distinct pairs sharing their past or future endpoint; J^4 collects chains $y_1 \prec x_1 = y_2 \prec x_2$ (MYZ's J^5, the chain $y_2 \prec x_2 = y_1 \prec x_1$, is J^4 with the two pairs relabelled, so the two are equal and absorbed in the factor 2); and J_{c}^6 is the connected part of the contribution of pairs with four distinct endpoints, which is non-zero only when the two intervals overlap. MYZ write all terms as integrals over configurations of points, with the joint distribution of the two interval cardinalities given by the volume of the intersection of the two intervals.

We evaluate the MYZ integrals by Monte Carlo with an importance sampler adapted to nearly null intervals. In four dimensions the intersection volume of two causal intervals has no closed form; we compute it by quadrature, using Green's theorem for the intersection of discs at fixed time and fixed coordinate along one axis. In two dimensions the prediction agrees with direct sprinklings at five densities $\rho V = 25$–400 ($|z| \le 1.56$, 4000 sprinklings each) and, for the smeared action with $\epsilon = 0.5, 0.25$, at ten more points ($|z| \le 1.72$). In four dimensions the agreement holds for $\rho V = 25, 50, 100$ (table 3 in appendix D), but the Monte Carlo error grows quickly with density (23% at $\rho V = 100$). This makes the full MYZ integrals impractical as a scaling tool, and in the rest of the paper we follow a different strategy: we compute *exactly* the terms that dominate asymptotically, and measure the remaining terms on sprinklings.

3 The same-pair term and the boost range

We now compute the term of (7) that dominates asymptotically, J^1, and show that it is controlled by the range of boosts that fits inside the region. Consider a nearly null pair, with small proper time τ between its endpoints. Its interval has volume $c_d\tau^d$ ($c_4 = \pi/24$), so its cardinality is a Poisson variable of mean $\mu = \rho c_d \tau^d$ that stays of order one however large the region is. What grows with the size of the region is the number of such pairs. Writing $y = x - \tau(\cosh\chi, \sinh\chi\,\hat{\boldsymbol{n}})$, the measure of the past hyperboloid is $\tau^{d-1}\sinh^{d-2}\chi\,\mathrm{d}\tau\,\mathrm{d}\chi\,\mathrm{d}\Omega$, and the pair fits inside the region as long as its rapidity satisfies $\tau e^{\chi}/2 \lesssim T(x,\hat{\boldsymbol{n}})$, where

$$T(x,\hat{\boldsymbol{n}}) = \sup\{\lambda :\ x - \lambda(1,\hat{\boldsymbol{n}}) \in M\} \tag{8}$$

is the extent, in time, of the past null ray from x in the direction $\hat{\boldsymbol{n}}$ that remains inside the region. The rapidity integral then gives a factor $(T/\tau)^{d-2}$, which is the volume of the available range of boosts. Inserting this in (4) and changing variables to μ gives, to leading order in ρV and for $d = 4$ (appendix C),

$$\Big\langle \sum_{y\prec x} h(n) \Big\rangle \simeq (\rho V)^{3/2}\,\frac{G/V^{3/2}}{8\sqrt{c_4}} \sum_{n\geq 0} h(n)\,\frac{\Gamma(n+\frac{1}{2})}{n!}\,, \qquad G = \int_M \mathrm{d}V_x \int_{S^2} \mathrm{d}\Omega\; T(x,\hat{\boldsymbol{n}})^2\,, \tag{9}$$

where $\mathrm{d}V_x$ is the physical volume element (in FLRW, $a(\eta_x)^4\mathrm{d}^4x$). In general dimension the exponent is $2 - 2/d$, with a weight $\Gamma(n + 2/d)/n!$. Two special cases are needed below.

Two dimensions. For $d = 2$ the boost volume is $\ln(T/\tau)$ and the weight is $\Gamma(n+1)/n! = 1$, so that $\langle \sum h\rangle = \rho V \ln\rho V \sum_n h(n) + O(\rho V)$. The same-pair term contributes $\sum_n f(n)^2$ ($= 24$ for the local action) to the coefficient of $\rho V \ln \rho V$ in the variance. In two dimensions the connected term J^6_{c} grows with the same logarithm (section 4), so the variance is Poissonian up to a logarithm, $(\Delta S)^2 \simeq b\,\rho V \ln \rho V$, with $b = (1 + r_2)\sum_n f(n)^2$ and r_2 the two-dimensional analogue of the remainder defined in section 4.

The cross term. For $h(n) = f(n)$ the leading term of (9) vanishes. For the local action in $d = 4$, $\sum_k a_k \Gamma(k+\frac{1}{2})/(k!\sqrt{\pi}) \propto -1 + \frac{9}{2} - 6 + \frac{5}{2} = 0$. This is the cancellation that makes the mean action finite in flat space, and by (5) it implies that $\mathrm{Cov}(N, F)$ grows only like ρV.

Therefore, if the correlation terms in (7) do not grow faster than J^1, the variance is dominated by

$$J^1 \propto (\rho V)^{2-2/d} \sum_n f(n)^2\,\frac{\Gamma(n+2/d)}{n!} \quad\Longrightarrow\quad \Delta S \propto (\rho V)^{1-1/d}, \tag{10}$$

that is, $\Delta S \propto (\rho V)^{3/4}$ in four dimensions. The region enters only through G, the integrated squared length of the past null rays: the boost range available to nearly null intervals. The exponent is a direct consequence of Lorentz invariance together with a cut-off set by the size of the region, the same non-compactness that makes the number of links unbounded in a sprinkling [19]; it cannot be changed by a finite rescaling of the density.

Should the correlation terms follow J^1? We have no proof that the other terms in (7) grow no faster than J^1, but there is a reason to expect it. The underlying claim is common to all of them: for a given nearly null pair, the measure of other pairs whose weights are correlated with it is finite, i.e. it does not itself grow with the boost range. For J^6_{c}, which only receives contributions from overlapping intervals, this is plausible: two nearly null intervals overlap appreciably only if they lie along a common null direction with comparable rapidity, so J^6_{c} should inherit the boost range once, like J^1. For J^2, J^3 and J^4, which are built from the row and column sums of the matrix of pair weights (appendix B), e.g. $R_y = \sum_x f(y,x)$, a necessary condition holds exactly: the mean of R_y has no boost-enhanced part, since its leading term is proportional to $\sum_n f(n)\Gamma(n+\frac{1}{2})/n!$, which vanishes by the cancellation just described. This removes the obvious way in which these terms could acquire a second factor of the boost range, but it is not sufficient, because J^2 involves the covariance of two pairs that share y, which does not cancel when both lie along nearly parallel null directions. We have not estimated these finite measures analytically. In two dimensions the claim is borne out, with an $O(1)$ coefficient that grows as ϵ decreases: the correlation terms share the logarithm of J^1 and r_2 is finite (section 4). The numerical tests of the next section probe it directly in four dimensions.

The approach to the asymptotic regime is slow. The exact J^1 of the local action in the four-dimensional diamond, computed from (4) without Monte Carlo, has a local exponent $\frac{1}{2}\,\mathrm{d}\ln J^1/\mathrm{d}\ln \rho V$ that decreases from 1.26 at $\rho V \approx 35$ to 0.96 at $\rho V \approx 570$, 0.88 at $\rho V \approx 2300$ and 0.77 at $\rho V \approx 3\times 10^5$. Exponents measured at small densities, such as the $\Delta S \propto (\rho V)^{1.2}$ we find for $\rho V \leq 100$, are pre-asymptotic. The slow approach is that of the leading correction to (9), which we derive in appendix C: the exact mean is the asymptotic one times $1-(\rho V)^{-1/2}(a_n \ln \rho V + b_n) + \ldots$, with

$$a_n = \frac{\pi}{\sqrt{c_4}\, G/V^{3/2}}\,\frac{n!}{\Gamma(n+\frac{1}{2})} \tag{11}$$

fixed by the measure of the hyperboloid and b_n dependent on the region.

4 Numerical tests

4.1 Method

For each region we compute J^1, $\langle F\rangle$ and $\mathrm{Cov}(N,F)$ exactly from (4) and (5), measure the total variance on independent Poisson sprinklings, and define the correlation *remainder*

$$r = \frac{\mathrm{Var}(F) - J^1}{J^1} = \frac{J^2 + J^3 + 2J^4 + J^6_{\mathrm{c}}}{J^1}\,, \tag{12}$$

the relative size of all correlation terms. The asymptotic exponent (10) holds if r tends to a constant. Our criterion is that the slope of a weighted linear fit of r against $\ln \rho V$ be compatible with zero at 4σ. A slope compatible with zero is a weak statement on its own, so we also quote the bound it implies on a residual power law, $r \propto (\rho V)^{\delta}$: to first order the slope equals $\delta\,\bar{r}$. Such a law would change the asymptotic exponent to $p = 3/4 + \delta/2$ once the correlations dominate. The 4σ threshold serves as a pass/fail criterion that guards against false alarms among many comparisons; the quantitative statement is the 2σ bound on δ. Each sprinkling is an independent sample of the ensemble, and the statistical error of a variance estimated from M sprinklings is $\mathrm{std}[(S-\bar{S})^2]/\sqrt{M}$. Efficient codes for the BD action of large sprinklings exist [25]; ours is available with this paper (see the data statement).

Methodological note. All pass/fail thresholds (4σ, the tolerance of every automated test of this work) were fixed before the corresponding runs; for the causal diamond, after a first set of five densities and before the larger ones. The points added after independent reviews of earlier drafts ($\rho V = 12\,800$ and $25\,600$ in the diamond, 6400 in the FLRW cones) and the combined fit of the diamond that includes them also had their criteria fixed in advance. The criteria that failed are reported where they arise: the original and the extended criteria for the smeared action (section 4.4), a bound on the memory of the action (section 5) and the reproduction of a published χ^2 (section 6).

4.2 Causal diamond

Figure 1 shows the four-dimensional diamond with the local action, for $\rho V = 25$ to 6400 (500 to 6000 sprinklings per density). The total variance follows the exact J^1 with a constant offset. The local exponent of the total variance decreases with ρV like that of J^1, from 1.24 ± 0.04 at $\rho V \approx 35$ to 0.72 ± 0.05 at $\rho V \approx 4500$ and 0.71 ± 0.07 between 12 800 and 25 600. The last two values lie 2.6σ and 1.5σ below those of J^1 (0.85 and 0.81), as a slowly decreasing r would produce; local exponents from neighbouring densities are noisy, however, and the bound on a decreasing r from the fit of the remainder, which covers this deviation, is the more robust statement.

On these sprinklings alone, $r = 0.317 \pm 0.014$ with slope $+0.006 \pm 0.009$. We then combined them with the independent sprinklings of the smeared-action runs of section 4.4 (at $\epsilon = 1$, $\rho V = 400$–6400) and with 500 new sprinklings at each of $\rho V = 12\,800$ and 25 600 (added later; see the methodological note). The remainder at these two densities is $r = 0.38 \pm 0.10$ and 0.19 ± 0.08, compatible with the earlier mean (0.6σ and -1.7σ), and over the whole range $r = 0.321 \pm 0.012$, i.e. $(\Delta S)^2 \simeq 1.32\, J^1$, with slope $+0.002 \pm 0.007$ (0.3σ; figure 1(c)). The fit has $\chi^2 = 16.0$ for 9 degrees of freedom (p-value 0.07): the points scatter about 30% more than their formal errors, as is visible between $\rho V = 800$ and 6400 in figure 1(c). Scaling the slope error by $\sqrt{\chi^2/\mathrm{dof}}$, the bound on a residual power law is $-0.05 < \delta < 0.06$ at 2σ ($-0.11 < \delta < 0.12$ at 4σ), that is, $p \leq 0.78$ (0.81); with the unscaled errors it would be $-0.04 < \delta < 0.05$. An exponent as low as the Poisson value would require $r \to -1$, correlations that cancel J^1; we see no trend in that direction.

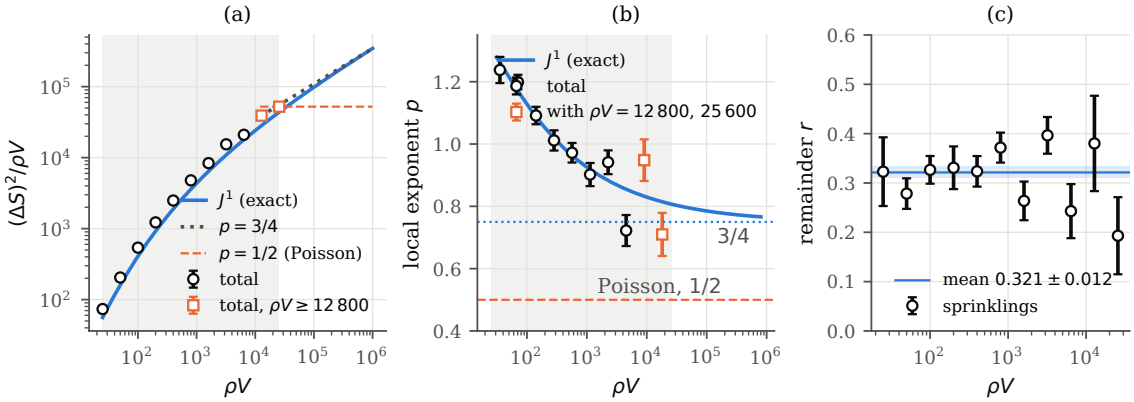


Figure 1. Local BD action in a four-dimensional causal diamond. (a) Total variance per element from sprinklings (points) and the exact same-pair term J^1 (line); the dotted line has the asymptotic slope of J^1, $p = 3/4$, and the dashed line the Poisson slope, $p = 1/2$; the squares are the independent sets at $\rho V = 12\,800$ and 25 600. (b) Local exponent of $\Delta S \propto (\rho V)^p$ for J^1 (line, exact) and for the total variance between consecutive densities (points, plotted at their geometric mean; the squares use the points at 12 800 and 25 600). The total follows J^1 towards 0.75; the Poisson value 0.5 is not approached. (c) Remainder $r = (\mathrm{Var}\, F - J^1)/J^1$ combining all independent sprinklings, including $\rho V = 12\,800$ and 25 600; the band is the weighted mean. The shading in (a) and (b) marks the simulated range; beyond it only J^1 is computed.

Three remarks on this fit. It combines two different but unbiased estimators of $\mathrm{Var}\, F$: for the first set, the variance of the full action minus the exact N contributions; for the others, the variance of F itself. Second, since the per-density errors, $\mathrm{std}(d^2)/\sqrt{M}$, could be underestimated for a heavy-tailed d^2, we checked them a posteriori with a bootstrap over sprinklings wherever the individual sprinklings were stored: the two errors agree within 3% in all 39 sets, so the excess scatter is not an underestimate of the individual errors. Third, the bound is stable against the fitted range: restricting the fit to $\rho V \geq 400$ gives $\delta < 0.04$ (2σ; 0.09 with the scaled error, $\chi^2 = 12.6$ for 5 degrees of freedom).

In two dimensions the logarithmic law of section 3 holds. For the diamond with $\rho V = 50$–$12\,800$, computed from the MYZ integrals (which agree with sprinklings wherever both are available, up to $\rho V = 3200$), the slope of $(\Delta S)^2/\rho V$ against $\ln \rho V$ is 24.43 ± 0.24 for the local action. Fitting each MYZ term separately over the same range, J^1 contributes 23.2, still approaching its asymptotic coefficient $\sum_n f(n)^2 = 24$, and the correlation terms, essentially J^6_c, contribute 1.76 ± 0.24. The coefficients do not add up ($23.2 + 1.76 \neq 24.43$) because J^1 is still curving towards its limit and the two fits weight the densities differently. Our estimate of the asymptotic coefficient is the asymptotic $\sum_n f(n)^2 = 24$ plus the correlation coefficient, $b \approx 25.8$, i.e. $r_2 \approx 0.07 \pm 0.01$, rather than $\sum_n f(n)^2$ alone. For the smeared action the correlations are larger, $r_2 = 0.42 \pm 0.04$ for $\epsilon = 0.5$ and 0.60 ± 0.06 for $\epsilon = 0.25$, a growth with decreasing ϵ similar to that of r in four dimensions (section 4.4). Binning the pairs that contribute to $\sum f^2$ by their rapidity $\eta = \frac{1}{2}\ln(\Delta u/\Delta v)$ gives a plateau for $|\eta| \lesssim 3$, cut off at $|\eta| \approx \frac{1}{2}\ln \rho V$: every unit of rapidity contributes the same, which identifies the logarithm with the boost range rather than with the boundary.

4.3 Past light cones and FLRW

Everpresent Λ involves the causal past of an observer, not a diamond. We repeat the analysis for the truncated past light cone $J^-(q) \cap \{t \geq 0\}$ in Minkowski spacetime and for the past light cone of an observer in a spatially flat FLRW spacetime, $\mathrm{d}s^2 = a(\eta)^2(-\mathrm{d}\eta^2 + \mathrm{d}\boldsymbol{x}^2)$, reaching the big bang at $\eta = 0$. We take $a = \eta^\alpha$ with $\alpha = 1$ (radiation) and $\alpha = 2$ (matter). In FLRW the interval volumes are computed exactly: at fixed η the section of a causal interval is a lens (the intersection of two balls), and the integral over η is done exactly by Gauss–Legendre quadrature in each polynomial piece. The boost-range integral of (9) becomes

$$G = \int \mathrm{d}V_x \int \mathrm{d}\Omega \int_0^{\eta_x} \frac{2\lambda\, a(\eta_x - \lambda)^4}{\sqrt{w}}\, \mathrm{d}\lambda, \qquad w = \frac{6}{\lambda^3}\int_0^\lambda a(\eta_y + s)^4\, s(\lambda - s)\, \mathrm{d}s, \tag{13}$$

where λ is the conformal extent of a nearly null interval along the null ray, $\eta_y = \eta_x - \lambda$ the conformal time of its past endpoint, and w is the average of a^4 over it, weighted by the profile $s(\lambda - s)$ of a thin four-dimensional diamond. For $a = 1$ it reduces to the Minkowski cone.

Table 1 summarises the results. In all regions the exact means converge to the asymptotic form (9), with deviations below 2.2% at $\rho V = 10^8$ that decrease monotonically with density, as

Table 1. Region dependence of the fluctuations of the local BD action in four dimensions. $G/V^{3/2}$ is the boost-range integral (9). The ratio $\langle N_k\rangle/\langle N_k\rangle_\infty$ of the exact mean to its asymptotic form (9) is evaluated at $\rho V = 10^8$ for $k = 0, \ldots, 3$, with exact means (diamond, Minkowski cone) or Monte Carlo means with errors below 0.2% (FLRW). The remainder r is the weighted mean over all simulated densities, including the points added later ($\rho V = 12\,800$, $25\,600$ in the diamond; 6400 in FLRW); the slope is that of r against $\ln \rho V$, with the χ^2 per degree of freedom of its fit.

Region	$G/V^{3/2}$	$\langle N_k\rangle/\langle N_k\rangle_\infty$ at 10^8	simulated ρV	r	slope of r	χ^2/dof
Diamond	1.4472(5)	0.995–0.987	25–25 600	0.321 ± 0.012	$+0.002 \pm 0.007$	1.8
Minkowski cone	0.8187	0.992–0.978	25–6400	0.342 ± 0.017	-0.001 ± 0.011	1.8
FLRW, $\alpha = 1$	1.3145	0.995–0.986	25–6400	0.346 ± 0.016	$+0.002 \pm 0.010$	0.8
FLRW, $\alpha = 2$	1.4194	0.997–0.981	25–6400	0.319 ± 0.015	-0.004 ± 0.010	0.7

$(\rho V)^{-1/2} \ln \rho V$. The prefactor G is also tested directly: the ratio of J^1 in the cone and in the diamond rises from 0.516 at $\rho V = 1.5 \times 10^5$ to 0.562 at 10^8, towards the predicted $0.8187/1.4472 = 0.566$. Fitting the exact means of the Minkowski cone at $\rho V = 10^6$, 10^7 and 10^8 with the form of section 3, the coefficient of the logarithm agrees with (11) within 2% for $k = 0, \ldots, 3$ (5.93 against 5.98 for $k = 0$). The remainder is a constant fraction of J^1 in all four regions, with values between 0.32 and 0.35; its slope against $\ln \rho V$ is compatible with zero in all cases (largest deviation 1.0σ); the diamond and the Minkowski cone show the excess scatter noted above ($\chi^2/\text{dof} \approx 1.8$). The FLRW points at $\rho V = 6400$, added later, give $r = 0.28 \pm 0.08$ ($\alpha = 1$) and 0.37 ± 0.09 ($\alpha = 2$), within 0.8σ of the earlier means. The two-dimensional analogues behave alike: in the Minkowski cone and in FLRW cones with $\alpha = 1, 2$ the total coefficient of $\rho V \ln \rho V$ is 24.09 ± 0.19, 24.26 ± 0.19 and 24.78 ± 0.24, against 24.43 ± 0.24 in the diamond. The big bang, where $a \to 0$, does not change the scaling. The exponent is universal; the geometry enters only through the amplitude, via G.

4.4 Smeared action

The smeared action is designed to suppress fluctuations, and it is the version that can give a phenomenologically relevant amplitude (section 5). For fixed ϵ the argument of section 3 applies unchanged, with $J^1_\epsilon \propto (\rho V)^{3/2} \sum_n f_\epsilon(n)^2 \Gamma(n + \frac{1}{2})/n!$, and J^1_ϵ is again computed exactly from (4). Whether the correlation terms remain proportional to it is less clear, because the smeared action receives contributions from intervals with up to $\sim 1/\epsilon$ elements, so the asymptotic regime requires $\epsilon \rho V \gg 1$.

We sprinkle the four-dimensional diamond at $\rho V = 400$–6400 (2000 sprinklings at the lowest density, 500 at the highest) and analyse each sprinkling with $\epsilon = 1, 0.5, 0.2, 0.1$; the decomposition described below uses a second, independent set of sprinklings of equal sizes. The exact means and covariances (5) agree with the simulation in all 40 comparisons ($|z| \leq 2.35$), and $\epsilon = 1$ reproduces the local result, $r = 0.345 \pm 0.025$.

Our pre-registered criterion for the smeared action was formulated for the remainder *after* subtracting the part of F linear in N (defined in the next paragraph), $r_\perp = r - \rho V g^2/J^1$. It was met for $\epsilon = 0.5$ (slope 0.9σ) but failed for $\epsilon = 0.2$ and 0.1, where $r_\perp$ grows with $\ln \rho V$ at 4.1σ and 6.7σ. The following analysis, designed after this failure, explains it.

For $\epsilon \leq 0.2$ two contributions of different origin compete at these densities. The part of F that is linear in N, of variance $\rho V\, g^2$ with $g = \mathrm{d}\langle F\rangle/\mathrm{d}(\rho V)$, is Poissonian, so it decays relative to J^1 as $(\rho V)^{-1/2}$; for $\epsilon = 0.1$ it falls from $2.75\, J^1$ at $\rho V = 400$ to $0.55\, J^1$ at 6400. To separate the contributions we measure every MYZ term directly on each sprinkling, using estimators built from the matrix $f(y, x)$ of pair weights and its row and column sums (appendix B). The generic MYZ integrator is not usable in this regime: at $\rho V = 6400$ its error on J^6_c was about ten times the signal even with 4×10^5 samples.

Figure 2 shows the result. As a check, the estimated J^1 agrees with the exact value in all fifteen cases ($|z| \leq 0.86$), and at $\rho V = 400$, $\epsilon = 0.1$, the estimated $J^4 = 17.86 \pm 0.06$ agrees with the independent MYZ Monte Carlo value 19.4 ± 2.1. For $\epsilon = 0.5$ all terms are flat, with $r \simeq 1.3$. For $\epsilon = 0.2$ and 0.1 the terms with a shared endpoint, which carry the Poissonian part, decrease relative to J^1, while J^6_c increases (from $-3.7\, J^1$ to $-0.1\, J^1$ for $\epsilon = 0.1$). The two trends compensate: the total remainder stays at $r \approx 3.0$ ($\epsilon = 0.2$) and $r \approx 4.5$ ($\epsilon = 0.1$), with slopes against $\ln \rho V$ of $+0.17 \pm 0.08$ and -0.29 ± 0.11, both below 3σ. The growth of $r_\perp$ that made the pre-registered criterion fail is therefore not a growing correlation: it is the approach of $r_\perp$ to r as the subtracted Poissonian term decays.

We test this explanation with a new criterion, adding 500 sprinklings at each of $\rho V = 12\,800$ and $25\,600$ (which extend $\epsilon \rho V$ to 2560) to both earlier sets: the slope of the unprojected r must be

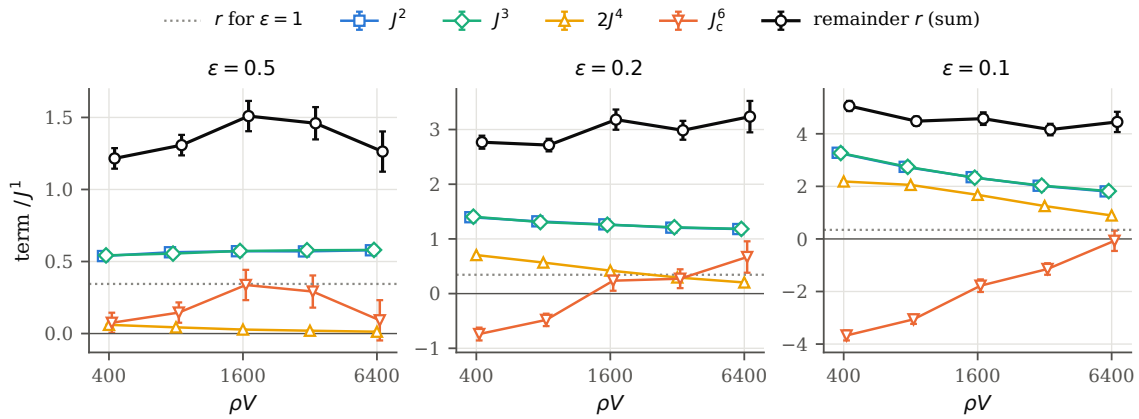


Figure 2. Decomposition of the correlation terms of the smeared BD action in a four-dimensional diamond, relative to the exact same-pair term J^1, for three values of the non-locality parameter. The terms with a shared endpoint (J^2, J^3, $2J^4$) decrease with ρV and the connected term J^6_c increases; their sum, the remainder r (black), stays approximately constant. Since J^6_c is obtained by subtraction, the error bars of r are those of the total variance. The vertical ranges differ between panels: for $\epsilon = 0.1$, J^6_c changes by $3.6\,J^1$. The decomposition into MYZ terms was measured only on the second of the two independent sets of sprinklings described in the text ($\rho V = 400$–6400). Points of different terms are shifted horizontally by a few % for clarity.

compatible with zero at 4σ, and the new points must be compatible with the earlier mean. At $\rho V = 25\,600$ we find $r = 1.41 \pm 0.16$, 3.33 ± 0.25 and 3.99 ± 0.32 for $\epsilon = 0.5$, 0.2 and 0.1, all compatible with the earlier means (0.5σ, 1.7σ, -1.9σ). The slopes over $\rho V = 400$–$25\,600$ are $+0.02 \pm 0.02$ (1.0σ), $+0.12 \pm 0.04$ (2.9σ) and -0.24 ± 0.06 (-4.1σ, $\chi^2 = 9.6$ for 5 degrees of freedom). The criterion is therefore met for $\epsilon = 0.5$ and 0.2 but *not* for $\epsilon = 0.1$, where r decreases. Much of this trend comes from the point at $\rho V = 400$, where $\epsilon\rho V = 40$ is presumably pre-asymptotic: for $\rho V \geq 800$ the slopes are $+0.01 \pm 0.03$, $+0.17 \pm 0.06$ (3.0σ) and -0.12 ± 0.07 (-1.6σ). The implied bounds on a residual power law at 2σ are $-0.02 < \delta < 0.05$, $0.01 < \delta < 0.07$ and $-0.08 < \delta < -0.03$. At $\rho V = 25\,600$ the bootstrap errors agree with the formula within 3%.

These trends have opposite signs and different consequences. For $\epsilon = 0.2$ the remainder may grow slightly; a power law with $\delta \leq 0.07$ would raise the asymptotic exponent to at most $p \approx 0.79$, away from the Poisson value. For $\epsilon = 0.1$ it decreases: the correlations grow more slowly than J^1. If the decrease saturates, or r decays to zero, $p = 3/4$ survives; if instead the linear trend in $\ln \rho V$ continued, the (negative) correlations would eventually cancel part of J^1. Within the accessible range, $\epsilon\rho V \leq 2560$, the smeared action is compatible with $\Delta S \propto (\rho V)^{3/4}$, but it has not reached a clean asymptotic regime: the individual terms are still drifting and the total shows weak trends of both signs. We cannot exclude a different behaviour for $\epsilon \to 0$ at fixed $\epsilon\rho V$; we return to this point in section 7.

5 Consequences for a fluctuating Λ

5.1 Amplitude

To connect the action with Λ we follow the logic of Everpresent Λ. The relation we adopt, equation (14), is a working hypothesis rather than a consequence of a causal-set dynamics; the comparison with Model 1 below only checks its normalisation. The BD action approximates the Einstein–Hilbert action, $S_{\rm EH}/\hbar = \kappa S$ with $\kappa = \ell^2/(8\pi\ell_p^2)$ in our normalisation, and the cosmological term is $\Lambda V/(8\pi\ell_p^2)$ [7]. Reading the action fluctuation in the past of the observer as a fluctuation of the cosmological term gives

$$\delta\Lambda\,\ell^2 = \frac{\Delta S}{\rho V}\,, \tag{14}$$

independently of ℓ_p. Here $\delta\Lambda$ is a standard deviation: the action fluctuation has zero mean, and, as in Model 1, it sets the rms of a zero-mean stochastic Λ. In Model 1 the fluctuating part of the action is Poissonian, $\Delta S_{\rm EH}/\hbar = \sqrt{N}/2$, and the discreteness length is $\ell^2 = \ell_p^2/(2\alpha)$ (equations 3.9 and 3.11–3.13 of [7]). Then $\kappa = 1/(16\pi\alpha)$, $\Delta S = 8\pi\alpha\sqrt{N}$, and (14) gives $\delta\Lambda\,\ell^2 = 8\pi\alpha/\sqrt{N}$, i.e. the $\delta\Lambda = 8\pi\alpha/\sqrt{V}$ of Model 1, as it must, since Model 1 defines its fluctuating term through the same relation.

A further question concerns boundaries. The mean action of a region with a boundary contains boundary and corner terms [11, 12, 22], which do not enter here because we use only the

fluctuation. The fluctuation itself, however, depends on the boundary through G: the number of nearly null pairs is set by where the past null rays leave the region. One may ask whether it should be read as a fluctuation of the bulk cosmological term. The pairs that produce it are not concentrated near the boundary: they start at points x throughout the region, the integrand $T(x,\hat{\boldsymbol{n}})^2$ of G is largest deep inside it, and in the two-dimensional diamond every unit of rapidity contributes equally (section 4). In this sense the fluctuation is a bulk effect whose range is set by the region, rather than a boundary term; but its reading as $\delta\Lambda$ remains an identification. With the results of section 4,

$$\delta\Lambda\,\ell^2 = A\,(\rho V)^{-1/4}, \qquad A^2 = (1+r)\,\frac{G/V^{3/2}}{8\sqrt{c_4}}\sum_n f(n)^2\frac{\Gamma(n+\frac{1}{2})}{n!}\,. \tag{15}$$

The volume of the causal past of a comoving observer today, back to the big bang in a Planck 2018 ΛCDM background [26], is $V \approx 6.4\times10^{242}\,\ell_p^4$ (computed by quadrature in conformal time). With the local action and $\ell=\ell_p$ this gives $\delta\Lambda/\Lambda_{\rm obs}\sim10^{62}$; matching the observed value would require $\ell\sim10^{27}$ m, larger than the Hubble radius. The local action is therefore excluded as a source.

For the smeared action we find numerically that $\sum_n f_\epsilon(n)^2\Gamma(n+\frac{1}{2})/n! \to 4.112\,\epsilon^{5/2}$ as $\epsilon\to0$, so $A\propto\epsilon^{5/4}=(\ell/\ell_k)^5$. The observed Λ is obtained for $\ell_k\approx2.6\times10^{-23}$ m with $\ell=\ell_p$ ($\epsilon\sim10^{-49}$), or $\ell_k\approx1.25\times10^{-22}$ m with $\ell=\ell_{\rm cs}=7.1\,\ell_p$, the discreteness length that corresponds to $\alpha=0.01$ in Model 1 ($\ell^2=\ell_p^2/2\alpha$). These values use $r=0.33$; since $\ell_k\propto(1+r)^{1/10}$, the larger $r\approx3$–4.5 found in section 4.4 for small ϵ raises them by at most 15%. This is below the bound $\ell_k\lesssim10^{-19}$ m on the non-locality scale of relativistic non-local field theories from 8 TeV LHC data [27], as quoted in [28], and inside the window 10^{-29}–2×10^{-22} m that proposed optomechanical experiments could probe [28]. It requires, however, a large hierarchy $\ell_k/\ell_p\sim10^{12}$, so the coincidence problem that Everpresent Λ was meant to solve reappears as a tuning of ℓ_k. Equation (15) also changes the *time* dependence: since $\delta\Lambda\propto V^{-1/4}\propto H$ in the matter era, the envelope $\delta\Lambda/3H^2$ behaves as a fluid with $w\approx-1/2$, negligible at recombination ($\sim4\times10^{-5}$ at $z=1100$ for the amplitude that matches the observed Λ today) and growing until today, whereas for Model 1 $w\approx0$.

5.2 Temporal correlation along the worldline

The quantity ΔS is a standard deviation over sprinklings, whereas $\Lambda(t)$ fluctuates in time along a single history. The bridge is the correlation of the action between the past cones of two instants of the same worldline: it turns the ensemble variance at each volume into the variance of a stochastic process along the worldline. These cones are nested, and since the region is causally convex every pair of the smaller cone is a pair of the larger one with the same cardinality. A single sprinkling of the largest cone gives the action of all the nested cones. We measure it for the local action in four dimensions, in the Minkowski cone and in the matter-dominated FLRW cone (4000 sprinklings each, $\rho V=200$, 500, 1250 and 3200). The action is close to a random walk, with a small positive memory: the correlation between the action and its increment is $\gamma=0.069\pm0.015$ (weighted mean of the six pairs of adjacent cones; no individual pair reaches 4σ). We parametrise the correlation between the actions S_i, S_j of two nested cones, $V_i<V_j$, as

$$\operatorname{corr}(S_i,S_j) = \left(\frac{\operatorname{Var}S_i}{\operatorname{Var}S_j}\right)^{\beta/2}, \tag{16}$$

where $\beta=1$ is a random walk with independent increments. The six pairs of cones give β between 0.88 and 0.97, with bootstrap errors of 0.03–0.09; combining them with their bootstrap covariance gives $\beta=0.93\pm0.02$ (Minkowski) and 0.92 ± 0.02 (FLRW). The distribution of the action over sprinklings is close to Gaussian at these densities (excess kurtosis 0.7 at $\rho V=200$, compatible with zero at 3200; below 0.2 in absolute value for the increments), which supports the Gaussian process used below. A pre-registered requirement that $|\gamma|<0.2$ for every pair of cones failed for one non-adjacent pair in each geometry ($\gamma=0.29\pm0.12$ and 0.23 ± 0.09, within 1σ of the bound); it does not affect the value of β.

6 Confronting the data

6.1 Stochastic model

Section 5 excludes the local action by its amplitude, so the test of this section concerns the strongly smeared action ($\epsilon\sim10^{-49}$). For that regime we assume that the exponent $q=1/4$ and the memory β measured for the local action carry over; neither has been measured in that regime (section 7).

We model the cosmological constant along the observer's worldline as

$$\Lambda(t) = \Lambda_0\,V(t)^{-q}\,s(t), \tag{17}$$

where $V(t)$ is the volume of the causal past and s is an Ornstein–Uhlenbeck process [29] in $\ln V$ with unit variance and $\mathrm{corr}(s_1, s_2) = (V_1/V_2)^c$, i.e. with relaxation rate c per unit of $\ln V$. Model 1 of [7] corresponds to $q = c = 1/2$ and $\Lambda_0 = 8\pi\alpha$. The BD action, with (14), corresponds to $q = 1/4$ and $c = 3\beta/4$, and we take $\beta = 0.9$ from the measurement of section 5. The background is integrated with feedback, $(H/H_{100})^2 = \omega_r a^{-4} + \omega_m a^{-3} + \Lambda/3$, with Λ in units of H_{100}^2 and V accumulated along the way. We set $\Lambda = 0$ before $z = 10^4$, start the process from zero and count realisations with $H^2 < 0$ as collapsed. This is the convention that best reproduces the completion fractions published for Model 1: 10 000, 3258 and 575 out of 10 000 for $\alpha = 0.005$, 0.01 and 0.012, against 9990, 3228 and 535 in [7]. The last two agree within binomial errors (0.6σ and 1.7σ); for $\alpha = 0.005$ we find no collapse where [7] report ten, a difference in rare early collapses that we have not traced (it may lie in their random-number or start-up conventions). These fractions are regenerated by the test suite.

We follow the per-seed method of [8]. Each seed fixes the random numbers of the process, and for each seed we minimise χ^2 over ω_m, the amplitude (α or Λ_0) and the supernova magnitude M_B. The minimisation over ω_m and the amplitude uses a 10×10 logarithmic grid ($\omega_m = 0.02$–0.8) with two local refinements, a final resolution of about 5% in ω_m; M_B is minimised analytically. We then refine the 50 best seeds of each run with a continuous (Nelder–Mead) minimisation. Because the CMB fixes ω_m to better than 1%, the refinement lowers the minima substantially when the CMB is included (by up to ~ 100 for the best seed), and all values quoted below are refined. A seed is *good* if $\chi^2 < \chi^2_{\Lambda\mathrm{CDM}} + 14.7$. In every run, the 50th-best grid seed lies further above this threshold than the largest gain that the refinement produced among the 50, so no seed outside the 50 could become good unless it gained more than all of them. The margin is that implied by [8], who count seeds with $\chi^2 < 1500$ against their ΛCDM value of 1485.3 for supernovae alone. Our own supernova-only ΛCDM fit gives 1452.0, with best-fit parameters that agree with [30]; we could not reconstruct the exact configuration behind 1485.3, so we always apply the margin relative to our own ΛCDM fit to the same data. The data are:

- Pantheon+SH0ES supernovae [31, 30, 32], with the standard cut ($z_{\mathrm{HD}} > 0.01$ or Cepheid host: 1657 of the 1701 light curves, 77 of them calibrators) and the full statistical and systematic covariance;
- DESI DR2 BAO [33]: 13 measurements of D_V/r_d, D_M/r_d and D_H/r_d at 7 redshifts, with their covariance;
- Planck 2018 distance priors [34] on R and l_A, with $\omega_b = 0.02236$, the sound horizon integrated over each history, and z_*, z_{drag} from the fits of [35, 36] anchored to their Planck values.

The pipeline reproduces the Planck 2018 sound horizons to 0.08% and R, l_A within 1σ, and the DESI value of Ω_m from BAO alone (0.2972 against 0.2975 ± 0.0086). The treatment of radiation is simplified (no massive neutrinos, fixed ω_r), which induces a systematic of order 0.1–0.2% in l_A. A variant in which l_A and r_d are rescaled so that the Planck best fit reproduces their published values exactly gives the same conclusions.

Distance priors assume standard early physics. The BD model satisfies this ($\Lambda/3H^2 \sim 4 \times 10^{-5}$ at $z = 1100$); Model 1 does not (~ 0.5), so we fit it to supernovae and BAO only. Its BAO term still uses the sound horizon integrated over each history, which takes the fitting formula for z_{drag} [36] outside its range of validity, so the BAO exclusion of Model 1 is less robust than that of the BD model, whose Λ is negligible before recombination. For completeness, with the CMB included its minimum is +359.

6.2 Results

With supernovae alone both models produce rare histories that fit as well as ΛCDM: 0.25% of good seeds for Model 1, consistent with the $\approx 0.3\%$ of [8], and 0.48% for the BD action. The good BD seeds fix $\ell_k \approx 3 \times 10^{-23}$ m, consistent with the value inferred from the amplitude alone. Both, however, prefer $\Omega_m \approx 0.5$.

Adding BAO changes the picture (table 2). Even without the CMB, none of the 10^4 seeds of either model is good (fraction below 3×10^{-4} at 95% confidence): $\Delta\chi^2_{\mathrm{min}} = +84$ for Model 1 and +23 for the BD action, with ΛCDM refitted to SN + BAO. For Model 1 this agrees with the full Planck analysis of [8], which found $\Delta\chi^2 > 150$; our route is independent and does not use the CMB. The best BD seed fits the supernovae better than ΛCDM (-15) but misses the BAO by +39: no history achieves simultaneously the shape of the distance–redshift relation required by the supernovae and the scale required by the BAO. Adding the CMB, which is applicable to the BD model because its Λ is negligible at recombination, raises the minimum to +74 (with $\beta = 0.9$), and

Table 2. Fits of stochastic $\Lambda(t)$ histories (10 000 seeds per row), after the continuous refinement of the 50 best seeds. $\Delta\chi^2$ is relative to the ΛCDM fit to the same data. The last column splits the $\Delta\chi^2$ of the best seed by probe. Model 1 is shown without the CMB, since the distance priors are not applicable to it; its BAO term relies on the drag-epoch fit outside its range of validity (section 6).

Model	good seeds	$\Delta\chi^2_{\rm min}$ refined (grid)	best seed: SN / BAO / CMB
SN + BAO ($\chi^2_{\Lambda\rm CDM} = 1488.2$)			
Model 1 ($q = c = \frac{1}{2}$)	0	+84 (+87)	+64 / +20 / –
BD, $\beta = 0.9$	0	+23 (+26)	−15 / +39 / –
SN + BAO + CMB ($\chi^2_{\Lambda\rm CDM} = 1504.0$)			
BD, $\beta = 0.5$	1	+14 (+45)	+4 / +16 / −7
BD, $\beta = 0.88$	0	+70 (+116)	+19 / +56 / −5
BD, $\beta = 0.9$	0	+74 (+124)	+20 / +59 / −5
BD, $\beta = 0.97$	0	+52 (+135)	+33 / +26 / −7
BD, $\beta = 1$	0	+56 (+154)	+35 / +28 / −7

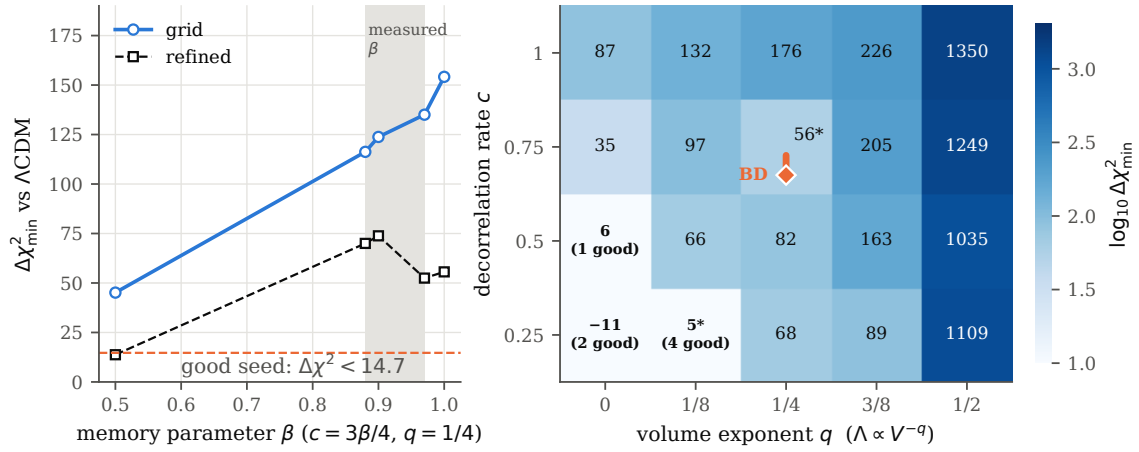


Figure 3. Robustness of the exclusion. Left: minimum $\Delta\chi^2$ with respect to ΛCDM for the BD dynamics ($q = 1/4$) as a function of the memory parameter β (10 000 seeds each), on the grid (circles) and after the continuous refinement of the 50 best seeds (squares); the orange line is the threshold for a good seed and the band marks the measured range of β. Right: refined minimum $\Delta\chi^2$ over the (q, c) family of Ornstein–Uhlenbeck dynamics, with the number of good seeds where there are any. Cells marked with an asterisk use 10^4 seeds (the cell $q = 1/4$, $c = 3/4$ is the BD action with $\beta = 1$ of the left panel); the others use 1000 seeds and are indicative (the 1000-seed minimum at $q = 1/4$, $c = 3/4$ is +116). The column $q = 1/2$, far above the threshold, shows grid values. The colour scale is clipped at $\Delta\chi^2 = 10$, so cells with smaller or negative minima share the lightest colour. Model 1 is the cell $q = c = 1/2$. The diamond marks the BD action ($q = 1/4$, $c = 3\beta/4$ with $\beta = 0.9$) and the bar the measured range of β.

none of the seeds that were good with supernovae alone survives. The conclusion does not depend on the threshold: with 25 or even 50 instead of 14.7 there are still no good BD seeds for any β in the measured range. Choosing the best of 10^4 seeds, each with its own fitted ω_m and amplitude, already favours the stochastic models; a comparison marginalised over seeds would penalise them further.

Figure 3 tests the robustness of this conclusion. Over the measured range of the memory parameter, $\beta = 0.88$–0.97, the minimum $\Delta\chi^2$ stays between +52 and +74. An unrealistically long memory, $\beta = 0.5$, does produce one good seed in 10^4 (+14), so the exclusion relies on the measured memory. Over the (q, c) family, good seeds appear only in the corner where Λ depends weakly or not at all on the volume and decorrelates slowly: $q = 0$ with $c = 1/4$ and $1/2$ (two and one good seeds in 1000, the best at −11) and $q = 1/8$, $c = 1/4$ (four in 10^4, minimum +5). The cells with the BD exponent $q = 1/4$ have no good seeds ($\geq +56$). The data therefore tolerate only a Λ that is close to a slowly wandering constant, with the rare good histories at a rate of order 10^{-3}–10^{-4}; the Model 1 exponent $q = 1/2$ is the worst column of the map ($\Delta\chi^2 > 1000$). This is distinct from the preference of DESI DR2 BAO, combined with CMB and supernova data, for an evolving dark-energy equation of state ($w_0 w_a$, at 2.8–4.2σ depending on the supernova sample [33]). That preference concerns a smooth drift of the mean; the models tested here are stochastic processes whose typical histories wander far more than the data allow.

7 Discussion

We have evaluated the fluctuations of the BD action that MYZ left as an open question. The central result is structural: in d dimensions the variance is dominated by the Poisson noise in the number of nearly null intervals, and that number is set by the volume of the boost group that fits inside the region; this gives $\Delta S \propto (\rho V)^{1-1/d}$. Only for $d = 2$ is the exponent $1/2$, and then with a logarithm. In four dimensions, for the local action, the exponent is $3/4$ within the simulated range in every region we tested, including the past cone of an FLRW observer reaching the big bang; the smeared action is compatible with it over a smaller range, with weak residual trends. If this persists asymptotically, a cosmological constant sourced by these fluctuations would scale as $\delta\Lambda \propto V^{-1/4}$, not as the $V^{-1/2}$ of Everpresent Λ, and a stochastic history built on this scaling and on the measured memory does not fit current data. Earlier cosmological tests of Everpresent Λ are those of [5, 37, 8]; to our knowledge this is the first confrontation of Everpresent-type dynamics with the DESI DR2 BAO data.

This does not refute Everpresent Λ as such, whose original argument relies on the Poisson fluctuation of N and not on the action. It shows that the action, the most natural dynamical quantity of the theory, does not provide the required mechanism. The obstruction is Lorentz invariance itself: any observable that counts intervals of fixed small volume inherits the boost range of the region. The same non-compactness biases the statistics of links in sprinkled causal sets [21]. A mechanism that yields $V^{-1/2}$ would have to suppress the contribution of highly boosted intervals, for instance through a boundary term or through a choice of frame that breaks local Lorentz invariance.

Four limitations qualify these conclusions.

1. *Asymptotics.* The exponent $3/4$ rests on one exact ingredient ($J^1 \propto (\rho V)^{3/2}$, verified up to $\rho V = 10^8$) and one empirical one: that the correlations remain a constant fraction of J^1. We verified the latter up to $\rho V = 25\,600$ in the diamond and 6400 in the past cones, with a residual power law bounded by $-0.05 < \delta < 0.06$ (2σ, errors scaled for the excess scatter). For the smeared action, up to $\epsilon\rho V = 2560$, the pre-registered test failed for $\epsilon = 0.1$, where the remainder decreases (-4.1σ; -1.6σ without the lowest density), and for $\epsilon = 0.2$ it grows slightly (2.9σ); the individual MYZ terms still drift. The physically relevant regime, $\rho V \sim 10^{242}$ with $\epsilon \sim 10^{-49}$, is an extrapolation over some 238 decades, which the heuristic argument of section 3 supports but does not prove; for the strongly non-local action that limit is, at present, a conjecture.

2. *Identification.* Equation (14) is an assumption, motivated by the cosmological term of the Einstein–Hilbert action and consistent with Model 1, but not derived from a dynamics for causal sets.

3. *Dynamics.* The data analysis uses a homogeneous $\Lambda(t)$ and the Ornstein–Uhlenbeck family (17). Spatial fluctuations of Λ are further constrained by the isotropy of the CMB [38]; non-Gaussian processes or a different coupling between the action and Λ were not explored. The temporal memory was measured for the local action only.

4. *Statistics.* The per-seed fits use a grid search followed by a continuous refinement of the 50 best seeds, rather than the MCMC of [8]. The refinement lowered the minima by up to ~ 100 when the CMB is included and produced good seeds where the grid had none, outside the measured BD dynamics; a still finer treatment could lower the minima further, although the BD minimum (+52 to +74) is far above the threshold. Our supernova-only ΛCDM χ^2 differs from that of [8] (section 6); all our margins are relative to our own fits.

Two results would settle what remains open. A calculation of the connected term J^6_c for $\epsilon \to 0$ at fixed $\epsilon\rho V$ would decide whether the exponent $3/4$ survives strong smearing, and a measurement of the temporal memory of the smeared action would remove the main assumption of the data test. If both confirm the behaviour of the local action, the conclusion is sharp: the BD action inherits the boost range of the region, and any mechanism for an everpresent Λ must find a way to discard it.

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Data availability
The code that produces every number and figure of this paper, together with its test suite (221 automated tests) and the saved intermediate results, is publicly available at https://github.com/carlosecheverriaarrondo/Conjuntos_causales and archived on Zenodo, https://doi.org/10.5281/zenodo.23413234. The supernova and BAO data are public and are downloaded, with checksum verification, by a script in the repository. Most tests check the code against exact results or fresh sprinklings; some re-read the saved results of the long runs, including the bootstrap check of the variance errors, and check the criteria quoted here. All statistical tolerances were fixed in advance, and each group of tests was checked to fail under a deliberately introduced error. A detailed log of every criterion, including those that failed, is part of the repository.

A Covariance with the number of elements

Let η be a Poisson process of intensity ρ on M and $F(\eta)$ a functional with finite second moment. The Mecke equation [24, theorem 4.1] gives $\mathbb{E}[NF] = \rho \int_M \mathbb{E}[F(\eta + \delta_x)]\,\mathrm{d}x$, while differentiating with respect to the intensity gives $\rho\,\mathrm{d}\mathbb{E}[F]/\mathrm{d}\rho = \rho \int_M \mathbb{E}[F(\eta + \delta_x) - F(\eta)]\,\mathrm{d}x$. Subtracting $\mathbb{E}[N]\,\mathbb{E}[F] = \rho V\,\mathbb{E}[F]$ from the first expression yields (5), a standard identity for Poisson functionals of Margulis–Russo type. For the local action, $F = \sum_k a_k N_k$ and $\langle N_k \rangle$ depends on ρ only through ρV, which leads to $\mathrm{Cov}(N, N_k) = (k+2)L_k - (k+1)L_{k+1}$. We evaluate $\mathrm{d}\langle F\rangle/\mathrm{d}\rho$ with a central difference and check it against this identity (relative error of $\mathrm{Cov}(N, F)$ below 10^{-7} for the local action at $\rho V = 25$–6400) and against sprinklings of the smeared action.

B Estimators of the MYZ terms

Smeared action. With thinned abundances the MYZ formalism carries over with three changes: the means become $\langle \tilde{N}_k \rangle = L_k(\epsilon\rho V)/\epsilon^2$; J^1 is no longer diagonal in k, because the two factors of a single pair are independent thinnings of the same interval; and in the joint distribution of two overlapping intervals the overlap is split into points kept by both (probability ϵ^2) and points kept by only one ($\epsilon(1-\epsilon)$ each), since thinning a Poisson process produces independent Poisson processes.

Per-sprinkling estimators. Let $f(y, x) = f(n(y, x))$ for $y \prec x$ and 0 otherwise, with row sums $R_y = \sum_x f(y, x)$ and column sums $C_x = \sum_y f(y, x)$. Sorting the terms of F^2 by shared endpoints gives, sprinkling by sprinkling,

$$\sum f^2 \to J^1, \quad \sum_y R_y^2 - \sum f^2 \to J^2, \quad \sum_x C_x^2 - \sum f^2 \to J^3, \quad \sum_z C_z R_z \to J^4, \tag{18}$$

and J^6_{c} is obtained by subtracting all of them from $(F - \langle F\rangle)^2$. The expectation of each statistic is the corresponding MYZ term, so the estimators are unbiased up to $O(1/M)$ from the sample mean. We checked the decomposition against an explicit enumeration of all 43^2 ordered pairs of the 43 related pairs of a sprinkling with 29 elements ($\rho V = 40$, $\epsilon = 0.3$); the two agree to a relative precision of 10^{-9}.

C The boost-range integral and its leading correction

Minkowski. Fix x and a unit vector $\hat{\boldsymbol{n}}$, and write the past point of a pair as $y = x - \tau(\cosh\chi, \sinh\chi\,\hat{\boldsymbol{n}})$. In four dimensions the measure is $\mathrm{d}V_y = \tau^3 \sinh^2\chi\,\mathrm{d}\tau\,\mathrm{d}\chi\,\mathrm{d}\Omega$ and the interval has volume $c_4\tau^4$. For $\chi \gg 1$ the point y lies at a distance $\lambda = \tau \mathrm{e}^{\chi}/2$ in time along the past null ray from x in the direction $\hat{\boldsymbol{n}}$, displaced from it by $O(\tau \mathrm{e}^{-\chi})$, so it lies in M if $\lambda < T(x, \hat{\boldsymbol{n}})$ up to corrections of relative order τ^2/T^2. Then

$$\int_0^{\chi_{\max}} \sinh^2\chi\,\mathrm{d}\chi = \frac{\mathrm{e}^{2\chi_{\max}}}{8} - \frac{\chi_{\max}}{2} + O(1) = \frac{T^2}{2\tau^2} - \frac{1}{2}\ln\frac{2T}{\tau} + O(1), \tag{19}$$

with $\mathrm{e}^{\chi_{\max}} = 2T/\tau$. The pairs with $\mathrm{e}^{\chi} = O(1)$ contribute only to the $O(1)$ term. With $\mu = \rho c_4 \tau^4$, $\int_0^\infty \tau\,P_n(\mu)\,\mathrm{d}\tau = \frac{1}{4}(\rho c_4)^{-1/2}\,\Gamma(n + \frac{1}{2})/n!$ and $\int_0^\infty \tau^3 P_n(\mu)\,\mathrm{d}\tau = \frac{1}{4}(\rho c_4)^{-1}$, where $P_n(\mu) = \mu^n \mathrm{e}^{-\mu}/n!$. Inserting the leading term in (4) gives (9):
$\langle N_n \rangle \simeq \rho^2 \int \mathrm{d}V_x \int \mathrm{d}\Omega\, \frac{\ell^2}{2}\, \frac{1}{4}(\rho c_4)^{-1/2}\Gamma(n + \frac{1}{2})/n! = \rho^{3/2}\, G\,\Gamma(n + \frac{1}{2})/(8\sqrt{c_4}\,n!)$. The logarithm gives the leading correction. Since $\ln(1/\tau) = \frac{1}{4}\ln\rho + O(1)$ for the values of τ that contribute, its ratio to the leading term is

$$-\frac{4\pi V \cdot \frac{1}{4}(\rho c_4)^{-1} \cdot \frac{1}{4}\ln\rho}{G \cdot \frac{1}{4}(\rho c_4)^{-1/2}\,\Gamma(n + \frac{1}{2})/n!} = -(\rho V)^{-1/2}\,\frac{\pi}{\sqrt{c_4}\,G/V^{3/2}}\,\frac{n!}{\Gamma(n + \frac{1}{2})}\,\ln\rho V + O\big((\rho V)^{-1/2}\big), \tag{20}$$

which is (11). The terms of order $(\rho V)^{-1/2}$ without the logarithm depend on $\int \mathrm{d}V\,\mathrm{d}\Omega\,\ln T$ and on the shape of the boundary, and are absorbed in b_n. In two dimensions $\int \mathrm{d}\chi = \chi_{\max}$ and the same steps give the logarithm of section 3.

FLRW. For $\mathrm{d}s^2 = a(\eta)^2(-\mathrm{d}\eta^2 + \mathrm{d}\boldsymbol{x}^2)$ we use conformal coordinates, in which the causal structure is that of Minkowski spacetime. With $\lambda = \tau_c \mathrm{e}^{\chi}/2$ the conformal extent of the pair along the null ray and τ_c its conformal proper time, the conformal measure becomes, for $\chi \gg 1$, $\tau_c^3 \sinh^2\chi\,\mathrm{d}\tau_c\,\mathrm{d}\chi\,\mathrm{d}\Omega \simeq \tau_c \lambda\,\mathrm{d}\tau_c\,\mathrm{d}\lambda\,\mathrm{d}\Omega$, and the physical measure carries the factor $a(\eta_y)^4 \simeq a(\eta_x - \lambda)^4$. The physical volume of the thin interval is $c_4 \tau_c^4\, w$, where w is the average of a^4 over it. A thin four-dimensional diamond of null extent λ has its volume distributed along the null direction with the profile $s(\lambda - s)$, which gives the w of (13). The integral over τ_c is the Minkowski one with c_4 replaced by $c_4 w$, i.e. $\frac{1}{4}(\rho c_4 w)^{-1/2}\Gamma(n+\frac{1}{2})/n!$, so that

$$\langle N_n \rangle \simeq \frac{\rho^{3/2}}{8\sqrt{c_4}}\,\frac{\Gamma(n+\frac{1}{2})}{n!}\int \mathrm{d}V_x \int \mathrm{d}\Omega \int_0^T \frac{2\lambda\, a(\eta_x - \lambda)^4}{\sqrt{w}}\,\mathrm{d}\lambda, \tag{21}$$

which is (9) with G given by (13) ($T = \eta_x$ for the cone that reaches the big bang). For $a = 1$, $w = 1$ and the λ integral is T^2, as in Minkowski spacetime.

D Validation of the MYZ integrals

Table 3 lists the comparisons between the variance predicted by the MYZ integrals, evaluated by Monte Carlo, and the variance measured on independent sprinklings of the flat causal diamond, quoted in section 2. In four dimensions the intersection volume of two intervals is computed by quadrature; at $\rho V = 100$ the Monte Carlo error of the prediction is 23%, so that point is a weak test.

Table 3. Variance of the BD action in the flat causal diamond: MYZ prediction against direct sprinklings (4000 per point in two dimensions, 6000 in four), with z the difference in units of the combined error.

	$d=2$, local			$d=4$, local		
ρV	prediction	sprinklings	z	prediction	sprinklings	z
25	638 ± 8	651 ± 17	0.64	1899 ± 82	1840 ± 97	−0.46
50	1968 ± 24	1971 ± 48	0.06	9775 ± 332	$10\,234 \pm 249$	1.10
100	5401 ± 68	5626 ± 126	1.56	$51\,224 \pm 12\,003$	$53\,932 \pm 1144$	0.22
200	$14\,099 \pm 177$	$14\,029 \pm 317$	−0.19			
400	$34\,734 \pm 468$	$36\,121 \pm 830$	1.46			
	$d=2$, $\epsilon=0.5$			$d=2$, $\epsilon=0.25$		
25	12.59 ± 0.27	13.36 ± 0.36	1.72	0.576 ± 0.015	0.595 ± 0.016	0.86
50	39.8 ± 0.9	40.5 ± 1.0	0.55	2.09 ± 0.05	2.10 ± 0.05	0.21
100	117.3 ± 2.7	113.4 ± 2.6	−1.03	6.46 ± 0.17	6.24 ± 0.15	−0.96
200	295 ± 8	299 ± 7	0.41	17.7 ± 0.5	17.2 ± 0.4	−0.74
400	733 ± 22	748 ± 17	0.56	43.4 ± 1.6	45.3 ± 1.0	0.98

References

[1] Bombelli L, Lee J, Meyer D and Sorkin R D 1987 *Phys. Rev. Lett.* **59** 521

[2] Surya S 2019 *Living Rev. Relativ.* **22** 5

[3] Sorkin R D 1997 *Int. J. Theor. Phys.* **36** 2759 (*Preprint* gr-qc/9706002)

[4] Sorkin R D 2007 *AIP Conf. Proc.* **957** 142 (*Preprint* 0710.1675)

[5] Ahmed M, Dodelson S, Greene P B and Sorkin R D 2004 *Phys. Rev. D* **69** 103523 (*Preprint* astro-ph/0209274)

[6] Ahmed M and Sorkin R D 2013 *Phys. Rev. D* **87** 063515 (*Preprint* 1210.2589)

[7] Das S, Nasiri A and Yazdi Y K 2023 *J. Cosmol. Astropart. Phys.* **10** 047 (*Preprint* 2304.03819)

[8] Das S, Nasiri A and Yazdi Y K 2024 *J. Cosmol. Astropart. Phys.* **10** 076 (*Preprint* 2307.13743)

[9] Benincasa D M T and Dowker F 2010 *Phys. Rev. Lett.* **104** 181301 (*Preprint* 1001.2725)

[10] Dowker F and Glaser L 2013 *Class. Quantum Grav.* **30** 195016 (*Preprint* 1305.2588)

[11] Buck M, Dowker F, Jubb I and Surya S 2015 *Class. Quantum Grav.* **32** 205004 (*Preprint* 1502.05388)

[12] Dowker F, Liu R and Lloyd-Jones D 2025 *Class. Quantum Grav.* **42** 205006 (*Preprint* 2501.00139)

[13] Moradi H, Yazdi Y K and Zilhão M 2025 *Class. Quantum Grav.* **42** 045017 (*Preprint* 2407.03395)

[14] Sorkin R D 2009 Does locality fail at intermediate length scales? *Approaches to Quantum Gravity* ed Oriti D (Cambridge: Cambridge University Press) pp 26–43 (*Preprint* gr-qc/0703099)

[15] Belenchia A, Benincasa D M T and Dowker F 2016 *Class. Quantum Grav.* **33** 245018 (*Preprint* 1510.04656)

[16] Aslanbeigi S, Saravani M and Sorkin R D 2014 *J. High Energy Phys.* **06** 024 (*Preprint* 1403.1622)

[17] Benincasa D M T, Dowker F and Schmitzer B 2011 *Class. Quantum Grav.* **28** 105018 (*Preprint* 1011.5191)

[18] Schmitzer B 2010 *Topology and curvature on causal sets* MSc dissertation Imperial College London

[19] Bombelli L, Henson J and Sorkin R D 2009 *Mod. Phys. Lett. A* **24** 2579 (*Preprint* gr-qc/0605006)

[20] Glaser L and Surya S 2013 *Phys. Rev. D* **88** 124026 (*Preprint* 1309.3403)

[21] Boguñá M and Krioukov D 2026 Sampling biases in sprinkled causal sets (*Preprint* 2610.03247)

[22] Machet L and Wang J 2021 *Class. Quantum Grav.* **38** 015010 (*Preprint* 2007.13192)

[23] Glaser L 2014 *Class. Quantum Grav.* **31** 095007 (*Preprint* 1311.1701)

[24] Last G and Penrose M 2017 *Lectures on the Poisson Process* (Cambridge: Cambridge University Press)

[25] Cunningham W J and Krioukov D 2018 *Comput. Phys. Commun.* **233** 123–133 (*Preprint* 1709.03013)

[26] Planck Collaboration 2020 *Astron. Astrophys.* **641** A6 (*Preprint* 1807.06209)

[27] Biswas T and Okada N 2015 *Nucl. Phys. B* **898** 113–131 (*Preprint* 1407.3331)

[28] Belenchia A, Benincasa D M T, Liberati S, Marin F, Marino F and Ortolan A 2016 *Phys. Rev. Lett.* **116** 161303 (*Preprint* 1512.02083)

[29] Uhlenbeck G E and Ornstein L S 1930 *Phys. Rev.* **36** 823

[30] Brout D *et al.* 2022 *Astrophys. J.* **938** 110 (*Preprint* 2202.04077)

[31] Scolnic D *et al.* 2022 *Astrophys. J.* **938** 113 (*Preprint* 2112.03863)

[32] Riess A G *et al.* 2022 *Astrophys. J. Lett.* **934** L7 (*Preprint* 2112.04510)

[33] DESI Collaboration 2025 *Phys. Rev. D* **112** 083515 (*Preprint* 2503.14738)

[34] Chen L, Huang Q G and Wang K 2019 *J. Cosmol. Astropart. Phys.* **02** 028 (*Preprint* 1808.05724)

[35] Hu W and Sugiyama N 1996 *Astrophys. J.* **471** 542

[36] Eisenstein D J and Hu W 1998 *Astrophys. J.* **496** 605

[37] Zwane N, Afshordi N and Sorkin R D 2018 *Class. Quantum Grav.* **35** 194002 (*Preprint* 1703.06265)

[38] Barrow J D 2007 *Phys. Rev. D* **75** 067301 (*Preprint* gr-qc/0612128)